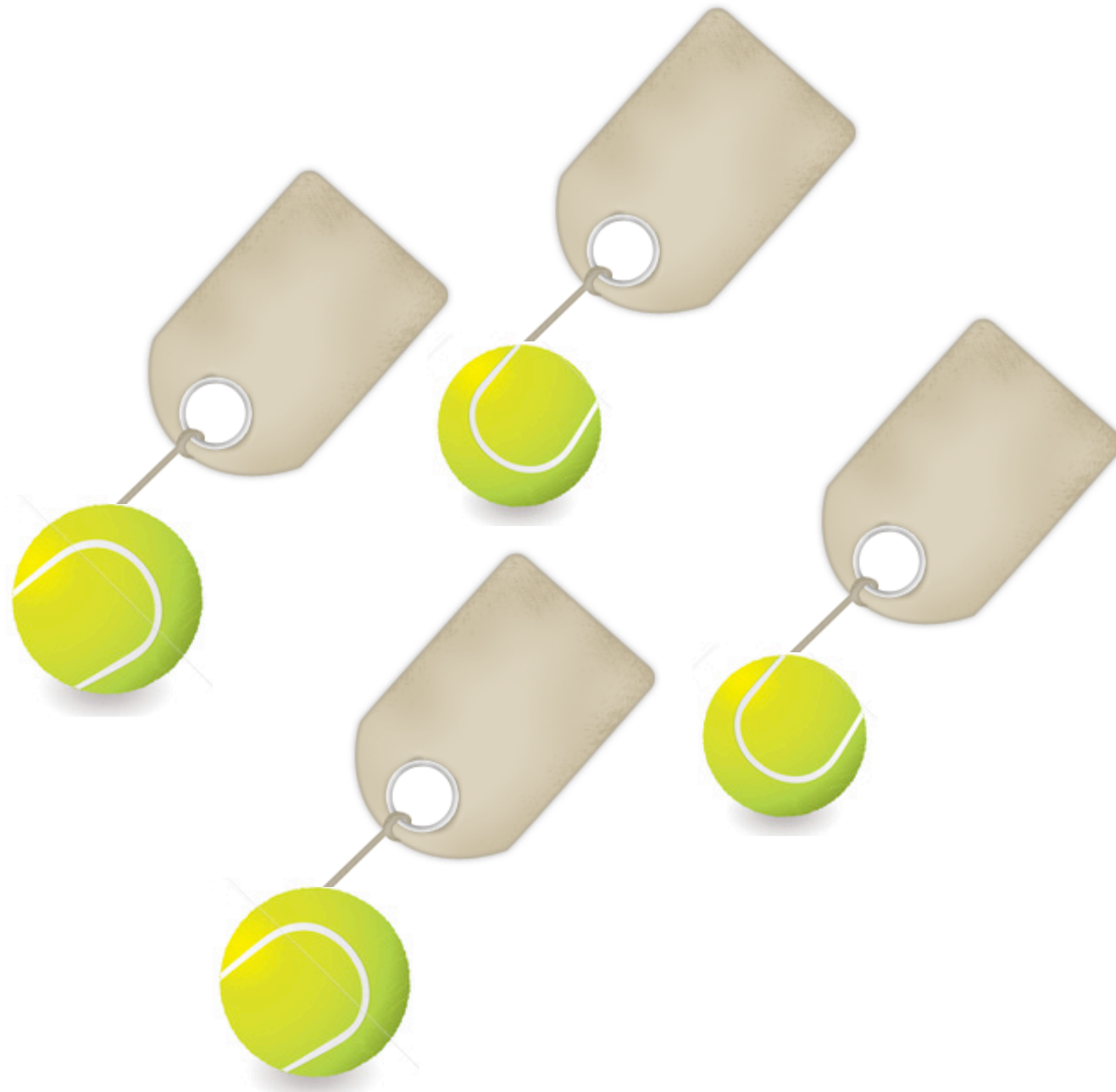


Dictionaries

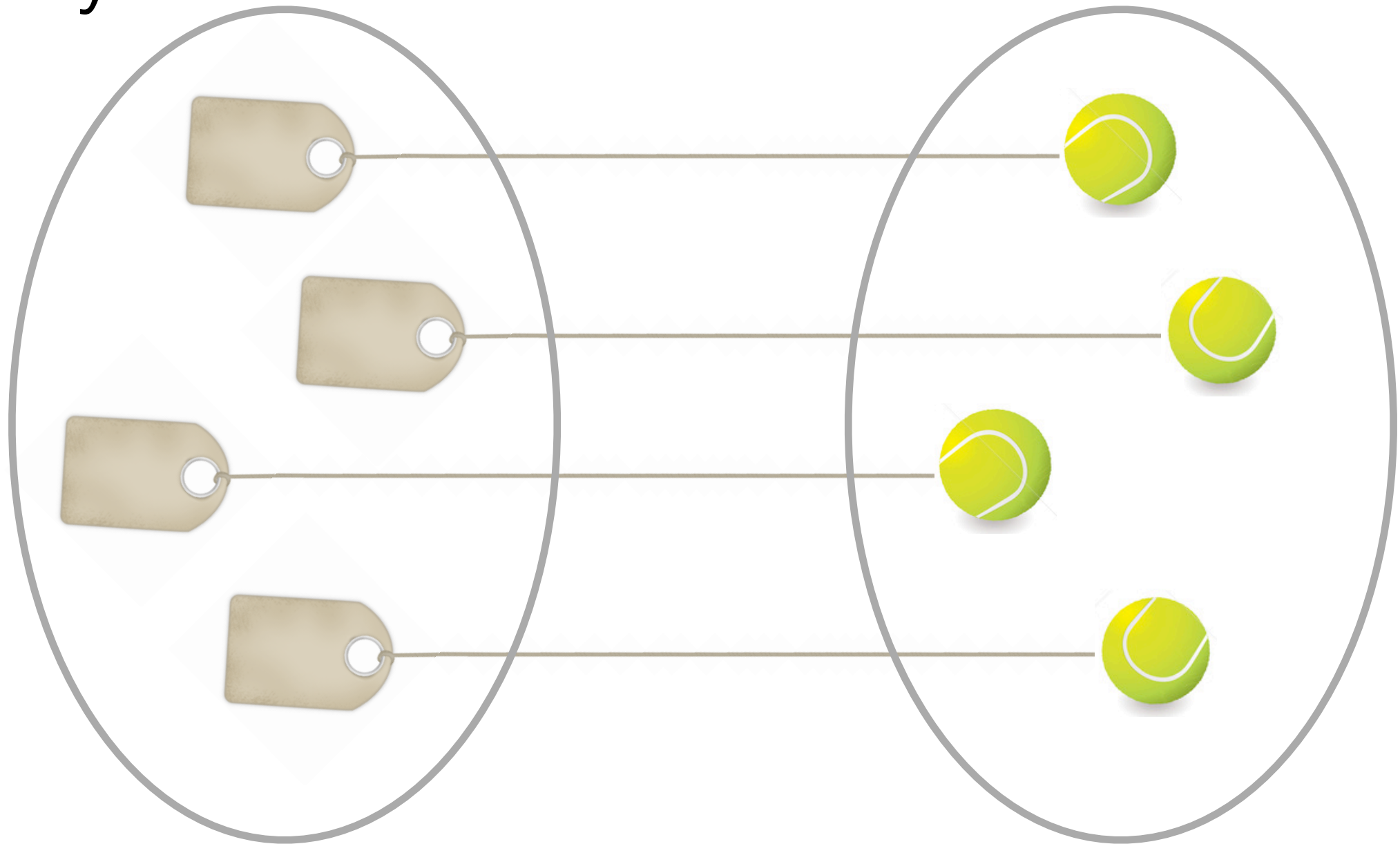
`dict`

Dictionary



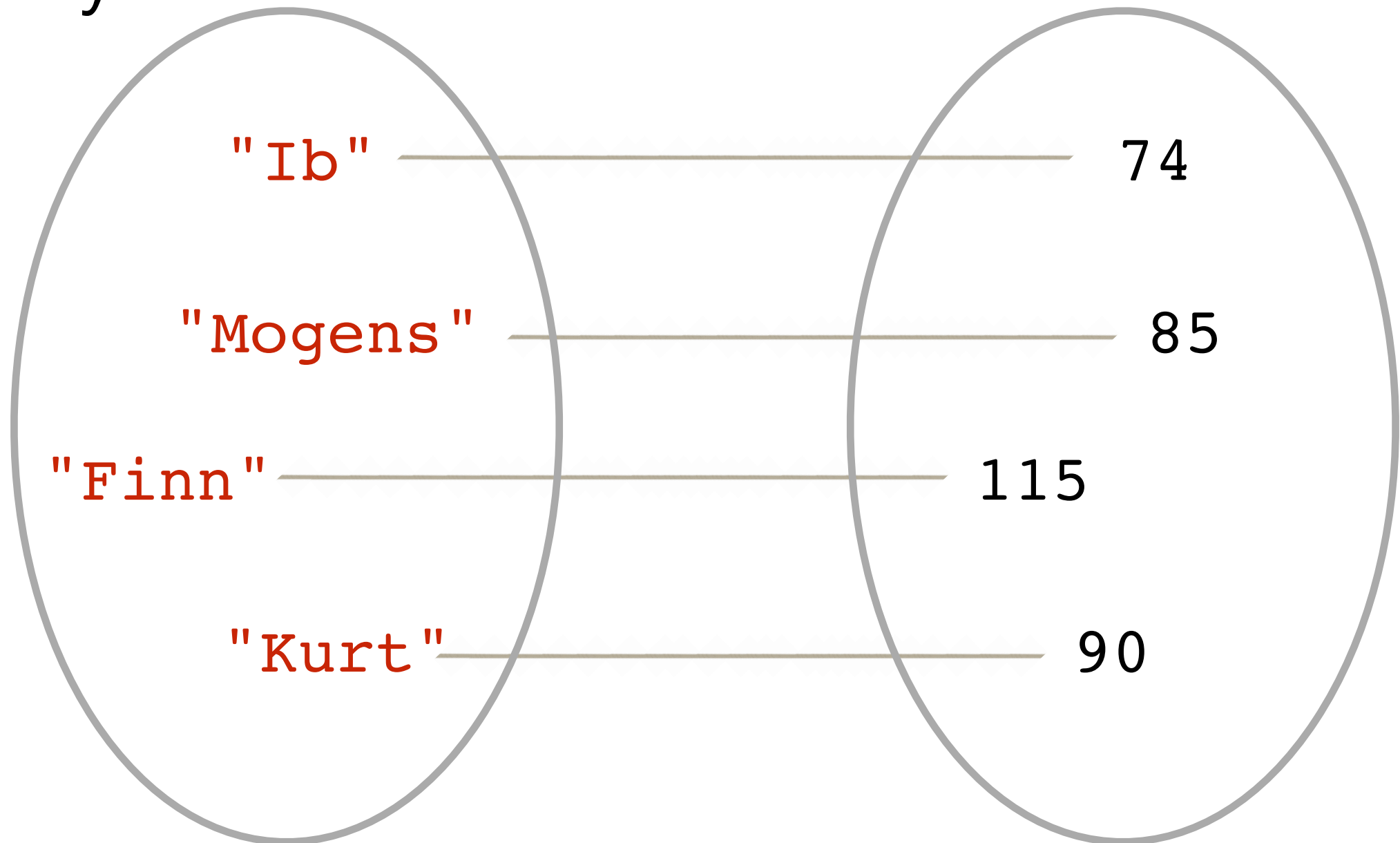
Keys

Values



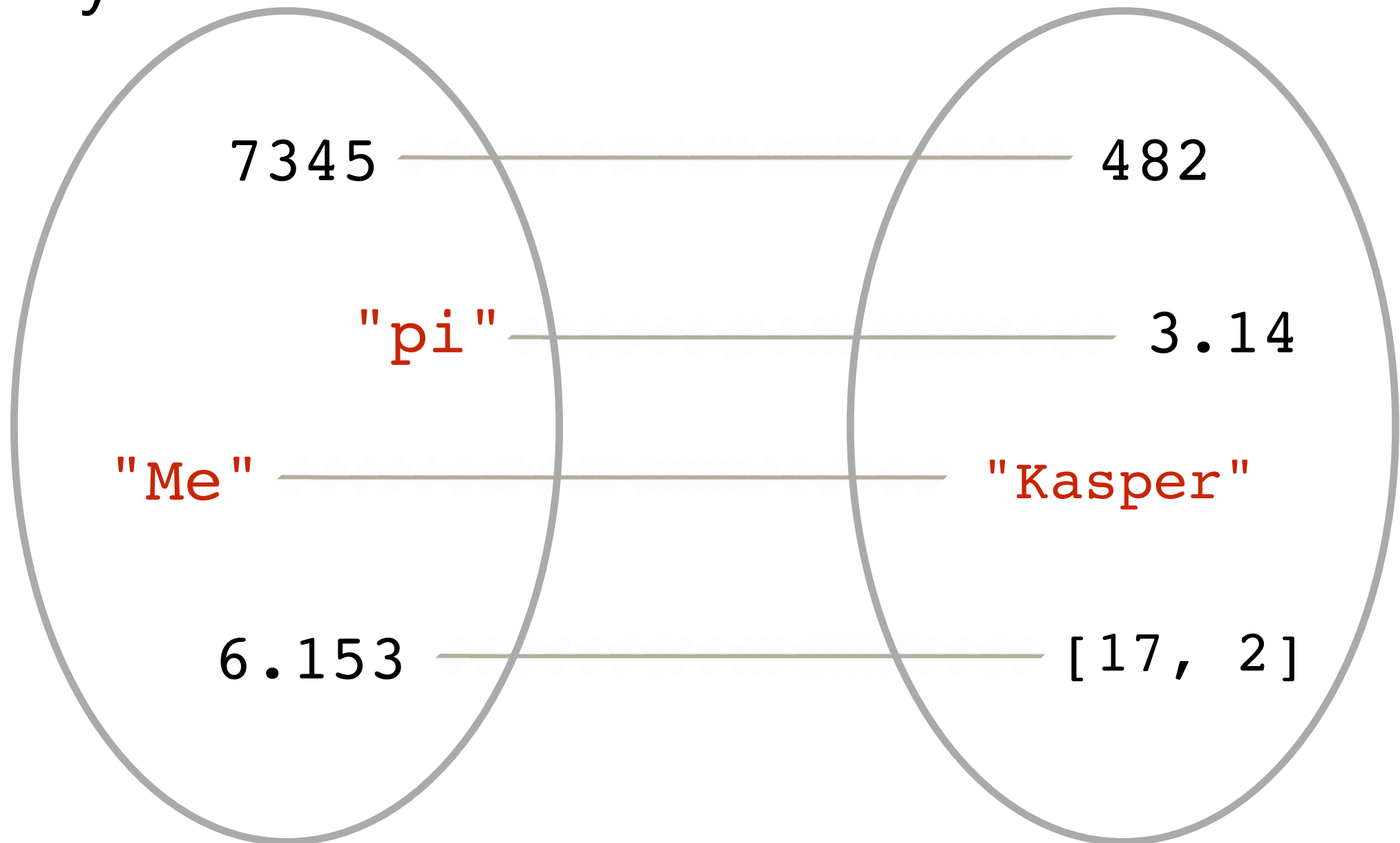
Keys

Values

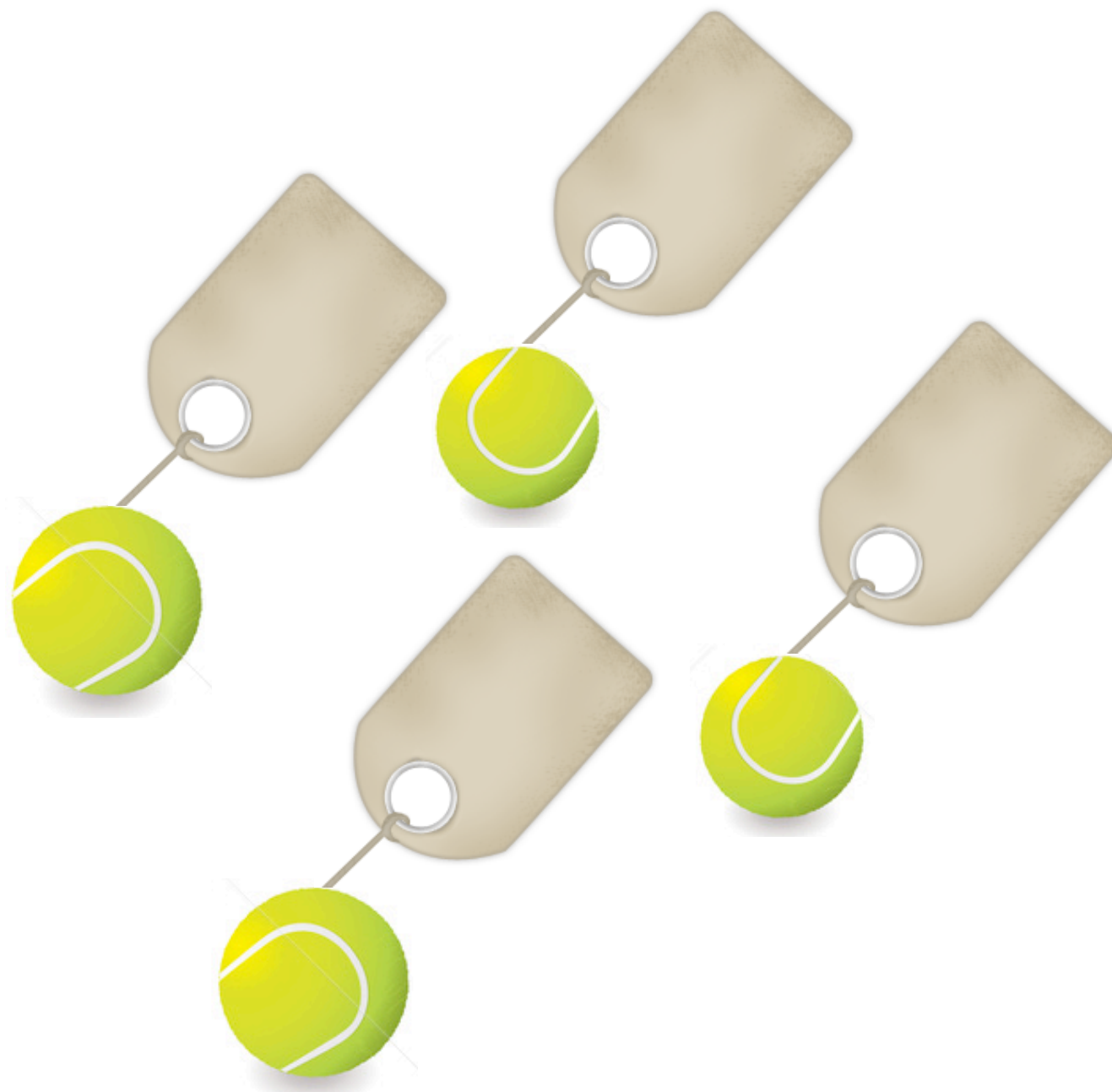


Keys

Values

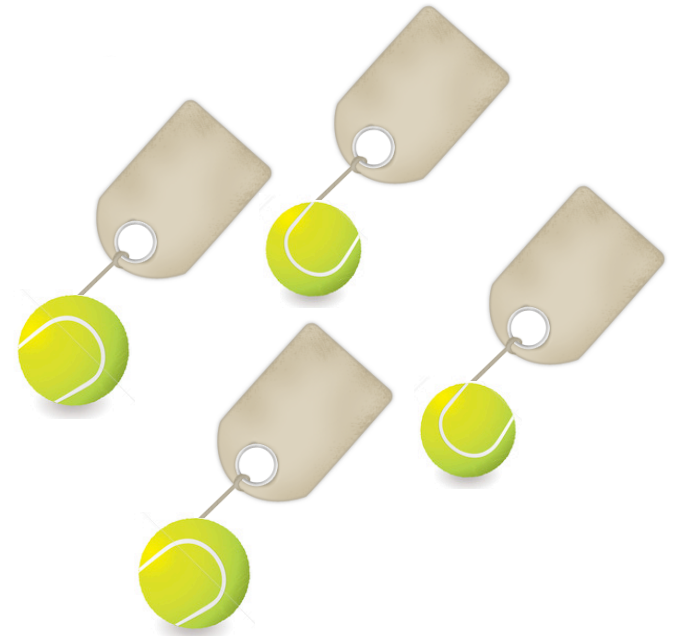


Hvad er et dictionarys værdi?



Dictionaries vs Lister

- “Database” vs. “Sekvens”
- Data er gemt i “key-value” par.
- Man tilgår en værdi vha. dens nøgle i stedet for et index



Definition af dictionaries

```
# new empty dictionary:  
d = {}
```

```
# new non-empty dictionary:  
d = {"bird": "flying animal",  
     "mouse": "small animal"}
```

Setting and getting values

```
d = {}
```

```
# assign a value to a key:
```

```
d["bird"] = "flying animal"
```

```
# get value for a key:
```

```
description = d["bird"]
```

```
# assign new value to a key:
```

```
d["bird"] = "living dinosaurs"
```

```
# delete key-value pair:
```

```
del d["bird"]
```

KeyError

```
animals = {"zebra": "striped animal",  
           "elephant": "large animal"}  
  
print(animals["Mogens"])
```

Test key membership

```
"bird" in my_dict
```

```
# True hvis "bird" er en key i my_dict
```

```
"bird" not in my_dict
```

```
# True hvis "bird" ikke er en key i my_dict
```

Patientdata

```
patient1_cpr = '170464-4361'  
patient1_data = {'height': 210, 'weight': 91, 'age': 57}
```

```
patient_age = patient1_data['age']
```

substitution



```
{ 'height': 210, 'weight': 91, 'age': 57 } [ 'age' ]
```

substitution



57

Patientdata

```
patient1_cpr = '170464-4361'  
patient1_data = {'height': 210, 'weight': 91, 'age': 57}  
  
patient2_cpr = '061246-4522'  
patient2_data = {'height': 162, 'weight': 65, 'age': 71}
```

Patientdata

```
patient1_cpr = '170464-4361'  
patient1_data = {'height': 210, 'weight': 91, 'age': 57}  
  
patient2_cpr = '061246-4522'  
patient2_data = {'height': 162, 'weight': 65, 'age': 71}  
  
database = {}
```

Hvordan kan vi konstruere en database for alle danskere, så vi kan hente patient data ud vha. et personnummer? F.eks:

```
patient_age = database['170464-4361']['age']
```

Patientdata

```
patient1_cpr = '170464-4361'  
patient1_data = {'height': 210, 'weight': 91, 'age': 57}
```

```
patient2_cpr = '061246-4522'  
patient2_data = {'height': 162, 'weight': 65, 'age': 71}
```

```
database = {}  
database[patient1_cpr] = patient1_data  
database[patient2_cpr] = patient2_data
```

```
{ '170464-4361': {'height': 210, 'weight': 91, 'age': 57},  
  '061246-4522': {'height': 162, 'weight': 65, 'age': 71} }
```

Patientdata

```
database = { '170464-4361': { 'height': 210, 'weight': 91, 'age': 57 },  
            '061246-4522': { 'height': 162, 'weight': 65, 'age': 71 } }
```

```
x = database[ '170464-4361' ][ 'age' ]
```

substitution

```
{ '170464-4361': { 'height': 210, 'weight': 91, 'age': 57 },  
  '061246-4522': { 'height': 162, 'weight': 65, 'age': 71 } } [ '170464-4361' ][ 'age' ]
```

substitution

```
{ 'height': 210, 'weight': 91, 'age': 57 } [ 'age' ]
```

substitution

57

Get the number of keys

```
my_dict = {"A": 3, "B": 7, "C": 4}

nr_keys = len(my_dict)

s = "There are {} keys in the dictionary"

print(s.format(nr_keys))
```

Under the hood

```
def len(x):  
    return x.__len__()
```

```
# Don't try this at home.  
# "__" means don't touch!
```

Dictionary methods

`d.keys()` `# alle keys`

`d.values()` `# alle values`

Tanketjek

```
x = 'Anna'
```

```
y = 'Lotte'
```

```
my_dict = {x: y}
```

```
y = 'Anders'
```

```
print(my_dict) # what is printed here?
```

```
z = my_dict[x]
```

```
z = 'Kirsten'
```

```
print(my_dict) # and here?
```

```
my_dict[x] = 'Georg'
```

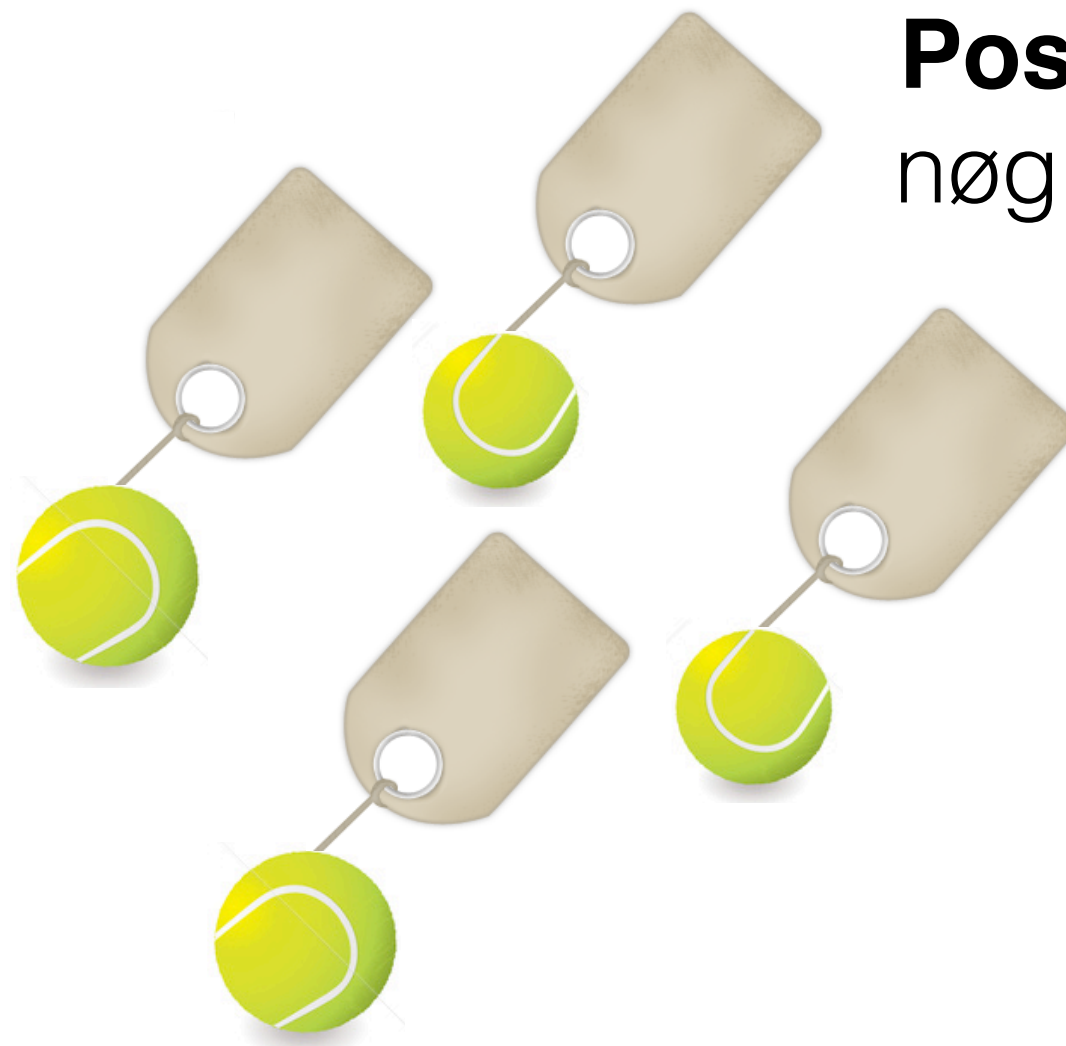
```
print(my_dict) # and here?
```

Værdier der holder værdier

Røret med
værdier

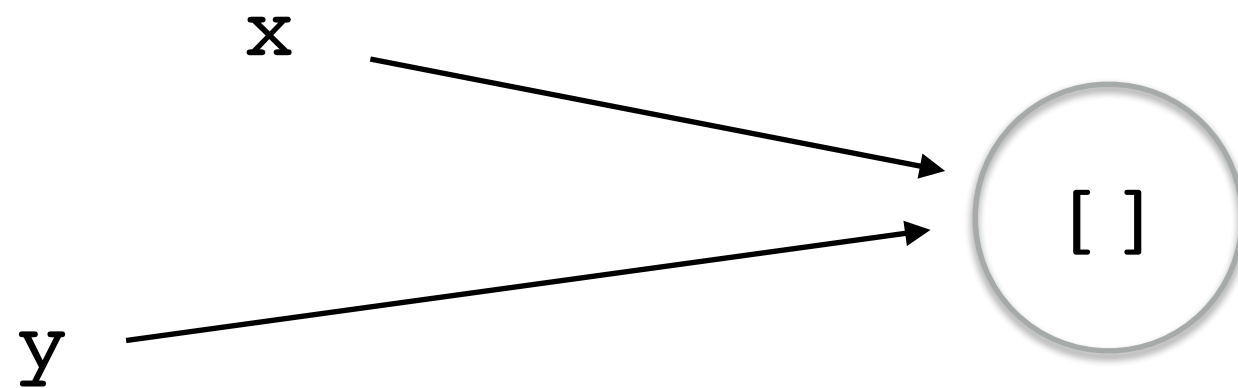
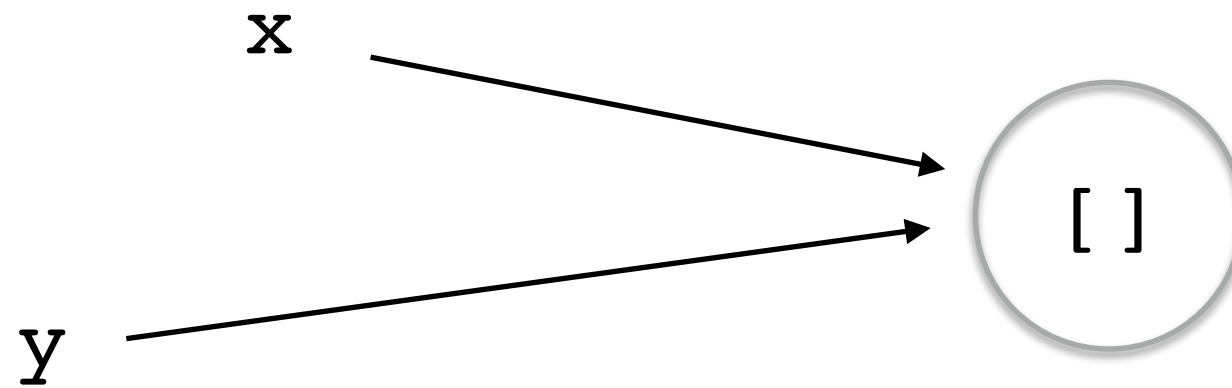
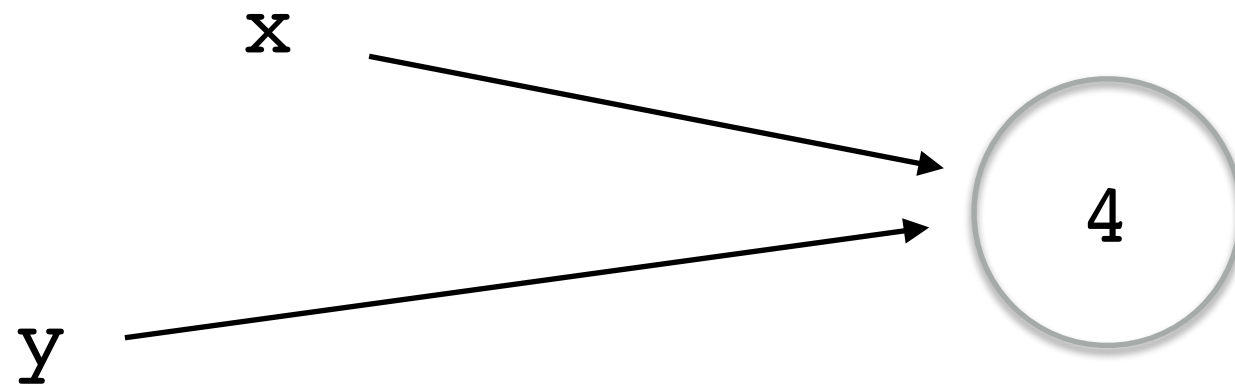


Posen med
nøgle-værdi
par



“Containers”

“aliasing” vs. “cloning”



Dictionaries

“aliasing” vs. “cloning”

```
a = { 'A' : 3, 'B' : 2 }
```

```
b = a # aliasing
```

```
b[ 'A' ] = 5
```

```
print(a)
```

```
a = { 'A' : 3, 'B' : 2 }
```

```
b = a.copy() # cloning
```

```
b[ 'A' ] = 5
```

```
print(a)
```

Lists

“aliasing” vs. “cloning”

```
a = [81, 82, 83]
```

```
b = a # aliasing
```

```
b[0] = 5
```

```
print(a)
```

```
a = [81, 82, 83]
```

```
b = a[:] # cloning
```

```
b[0] = 5
```

```
print(a)
```

Tanketjek

```
some_list = ['A', 'B', 'C']  
print(some_list) # what is printed here?
```

```
other_list = some_list[:]  
print(other_list) # and here?
```

```
other_list[2] = 'Mogens'  
print(other_list) # and here?
```

```
print(some_list) # and here?
```

Tanketjek

```
some_dict = {"A": 3, "B": 7, "C": 4}  
print(some_dict) # what is printed here?
```

```
other_dict = some_dict  
print(other_dict) # and here?
```

```
other_dict["B"] = 'Mogens'  
print(other_dict) # and here?
```

```
print(some_dict) # and here?
```